

Data Structures Lab – Assignment 9

April 08, 2026

Assignment: Trees

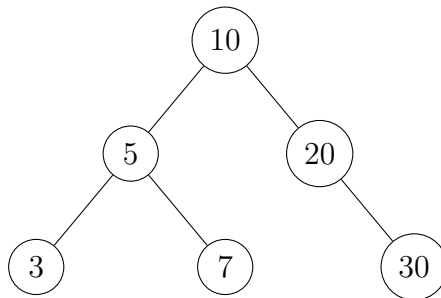
Q1. Binary Tree Traversals

Description: A binary tree is a hierarchical data structure where each node has at most two children (left and right). Tree traversal means visiting all nodes in a specific order.

Task: Write a program to perform:

- Inorder Traversal (Left, Root, Right)
- Preorder Traversal (Root, Left, Right)
- Postorder Traversal (Left, Right, Root)

Input:



Output:

Inorder: 3 5 7 10 20 30

Preorder: 10 5 3 7 20 30

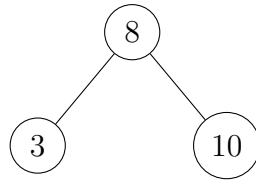
Postorder: 3 7 5 30 20 10

Q2. Search in Binary Tree

Description: Searching in a binary tree involves checking whether a given value exists in any node of the tree.

Task: Write a program to search for a given key in a binary tree.

Input:



Key = 10

Output:

Element found

Q3. Insertion in Binary Search Tree (BST)

Description: A Binary Search Tree (BST) is a binary tree where:

- Left subtree contains values smaller than the root
- Right subtree contains values greater than the root

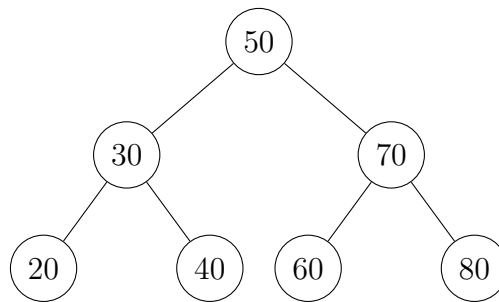
Task: Insert elements into a BST and print the inorder traversal.

Input: Insert: 50, 30, 70, 20, 40, 60, 80

Before Insertion:

Empty Tree

After Insertion:



Output:

20 30 40 50 60 70 80

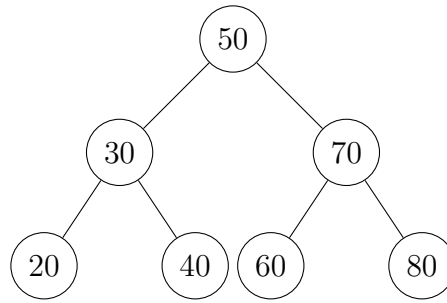
Q4. Deletion in Binary Search Tree

Description: Deletion in BST has three cases:

- Node is a leaf
- Node has one child
- Node has two children (replace with inorder successor/predecessor)

Task: Delete a node from a BST and print inorder traversal.

Input:



Delete = 50

Output:

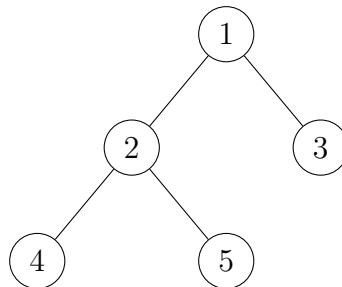
20 30 40 60 70 80

Q5. Lowest Common Ancestor (LCA)

Description: The Lowest Common Ancestor (LCA) of two nodes in a binary tree is defined as the deepest node that has both nodes as descendants (a node can be a descendant of itself).

Task: Write a program to find LCA of two nodes in a binary tree.

Input:



Nodes = 4, 5

Output:

LCA = 2

Explanation:

- Node 1 has descendants: 2, 3, 4, 5 → contains 2, 3, 4 and 5
- Node 2 has descendants: 4, 5 → contains both 4 and 5
- Node 4 → only contains itself ×
- Node 5 → only contains itself ×

Among the valid nodes (1 and 2), we select the **lowest (deepest)** node.

LCA = 2

Special Case: One Node is Ancestor of the Other

Example: Find LCA of 2 and 4

- Node 2 has 4 as a descendant
- Node 2 is also considered its own descendant

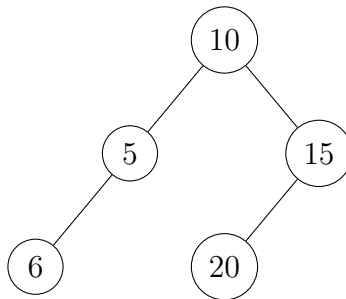
$$\boxed{\text{LCA}(2, 4) = 2}$$

Q6. Validate Binary Search Tree

Description: A valid BST must satisfy the property that all nodes in the left subtree are smaller than the root and all nodes in the right subtree are greater than the root, recursively.

Task: Write a program to check whether a binary tree is a valid BST.

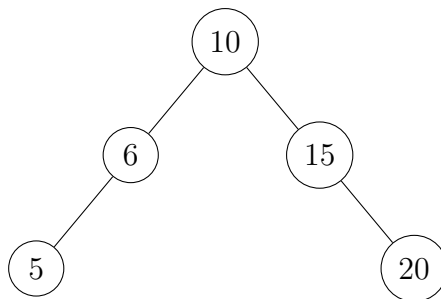
Input 1:



Output 1:

Not a valid BST

Input 2:



Output 2:

Valid BST